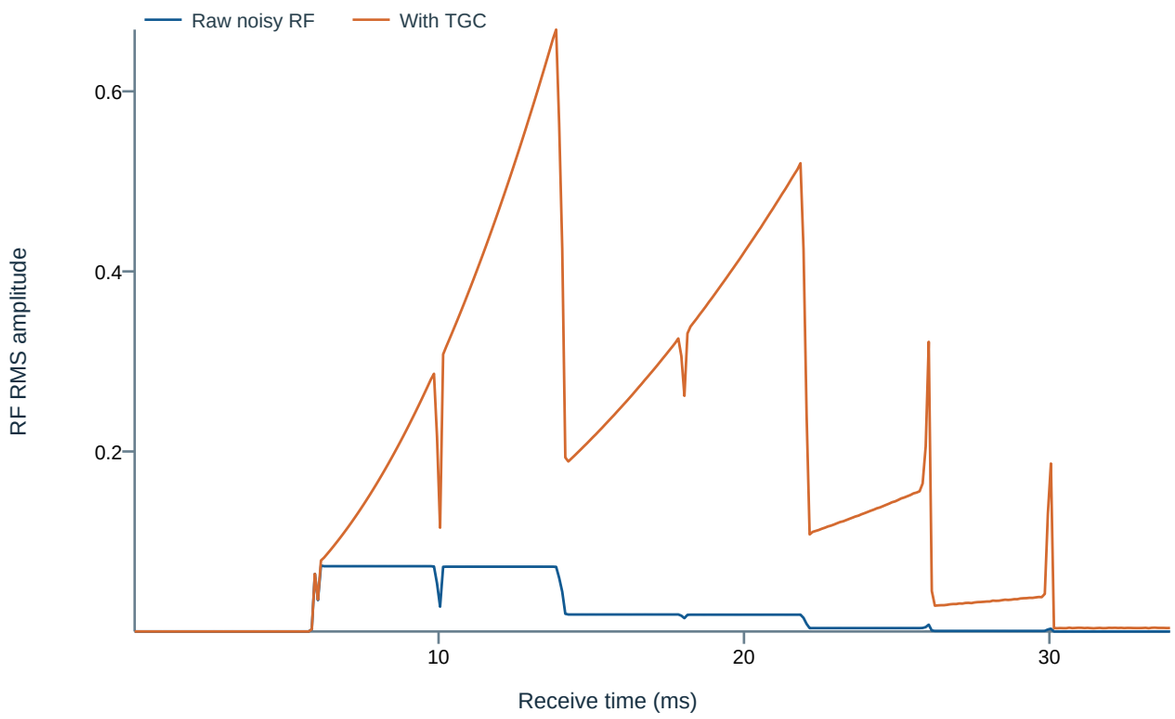
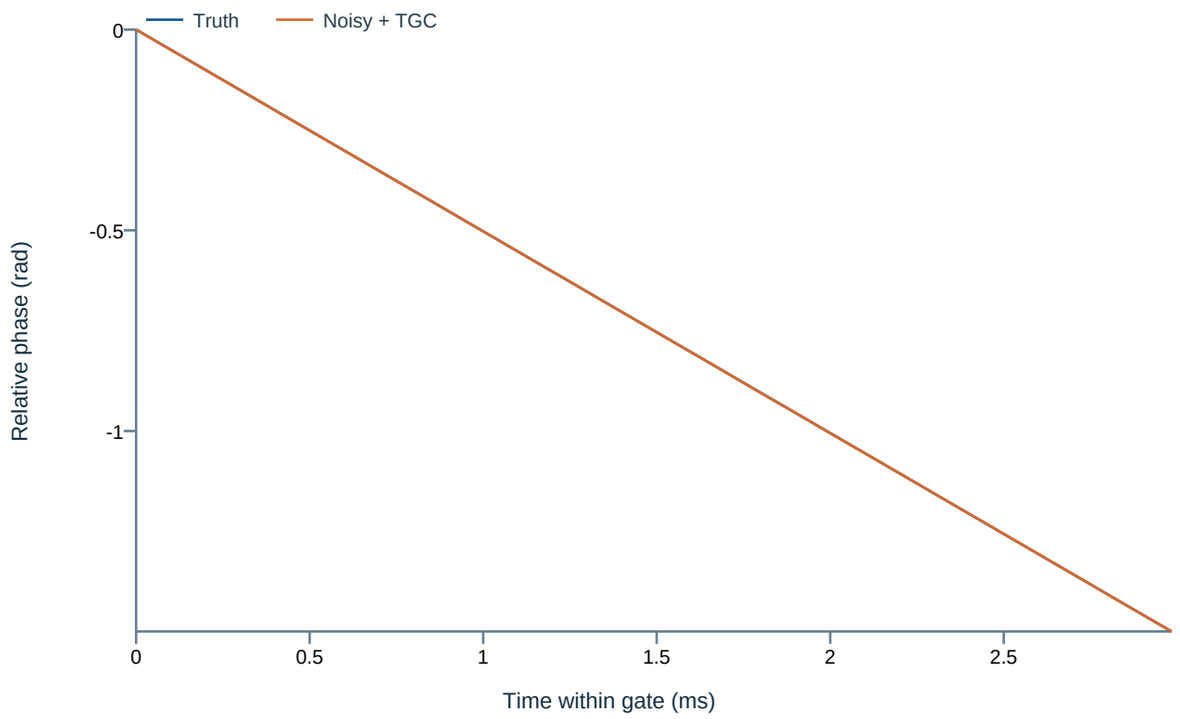
1 / Known-answer full-RF validation

6 controlled scattering layers | 1440 scatterers | Same 300 kHz carrier, 4 ms TX and 0.5 ms lag

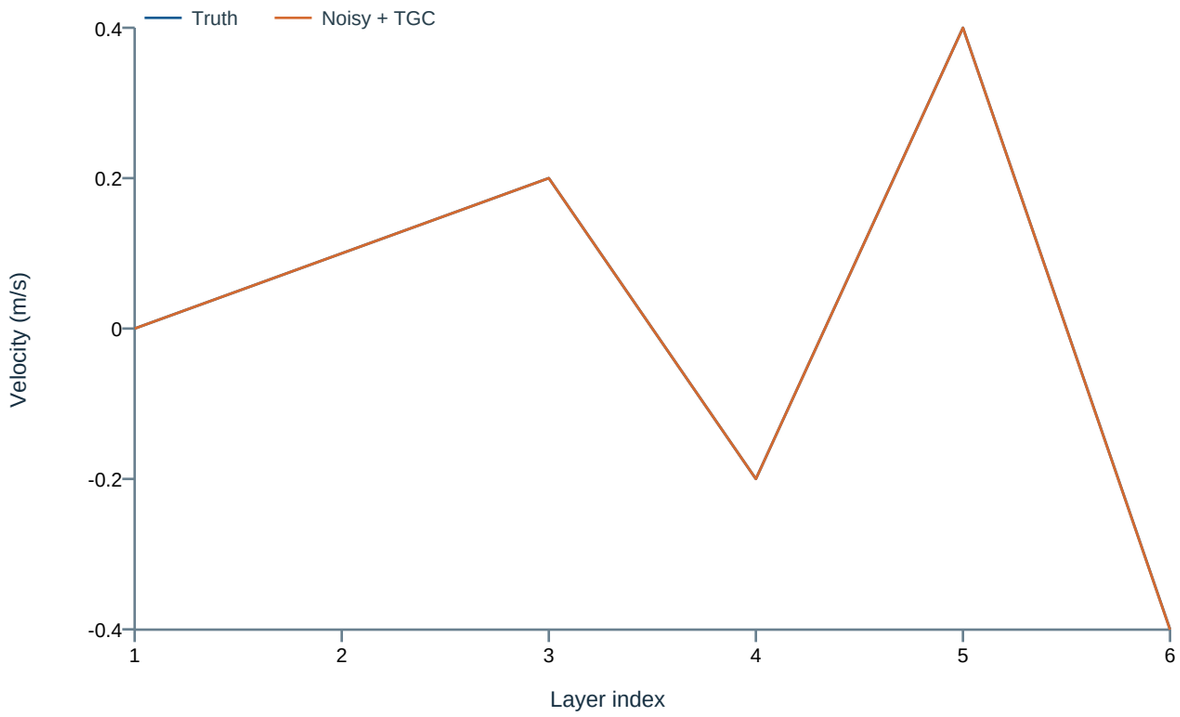
A. All layers are synthesized before selecting a gate



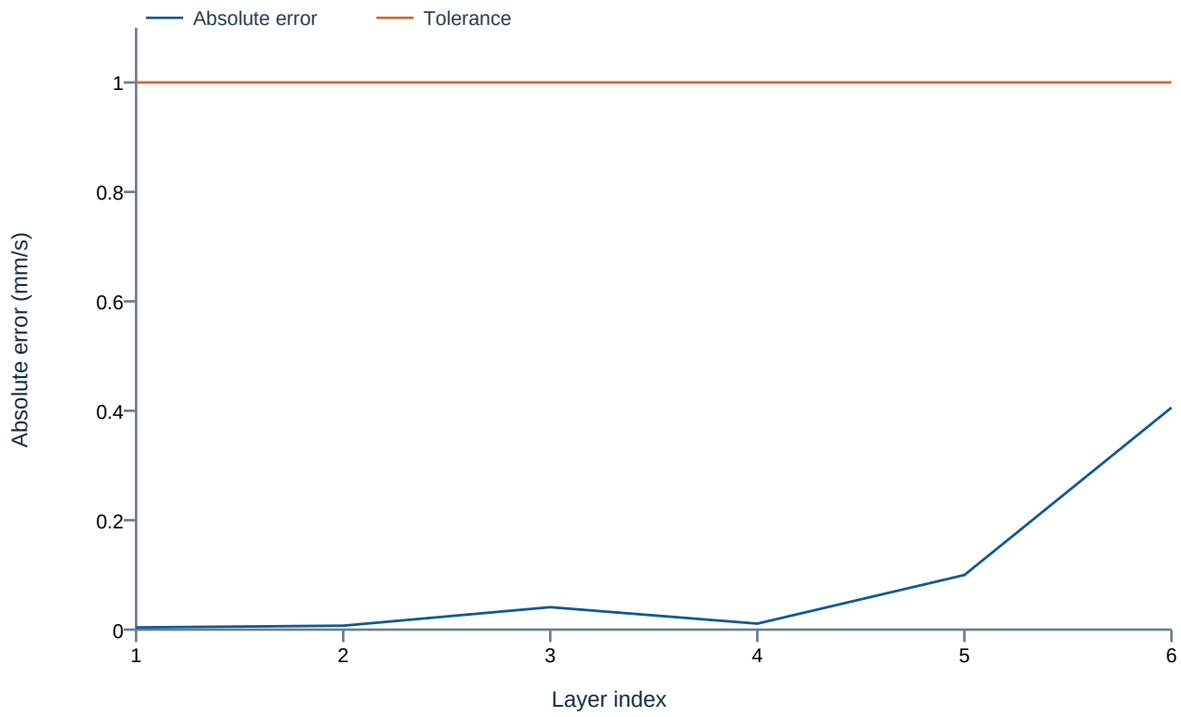
B. Selected gate: phase follows the known Doppler



C. Zero, positive and negative velocities all match



D. All six noisy gates pass the 1 mm/s criterion

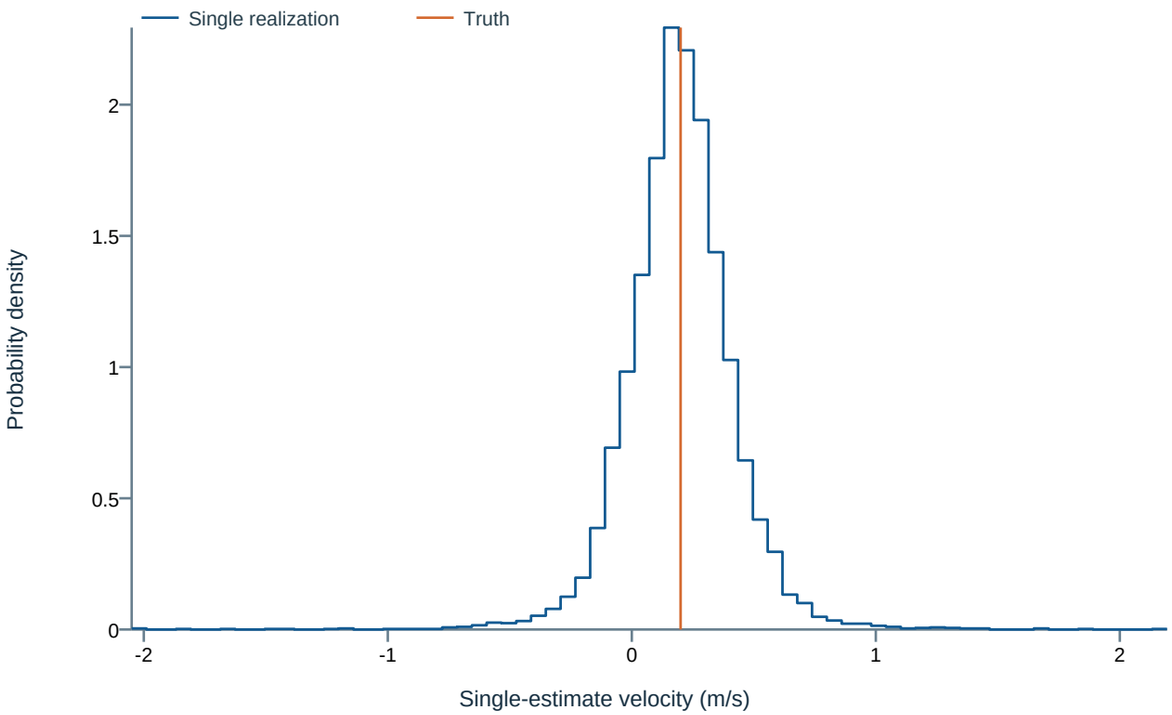


Narrowband teaching simulation | Positive radial velocity = away | Offline FIR delay compensated

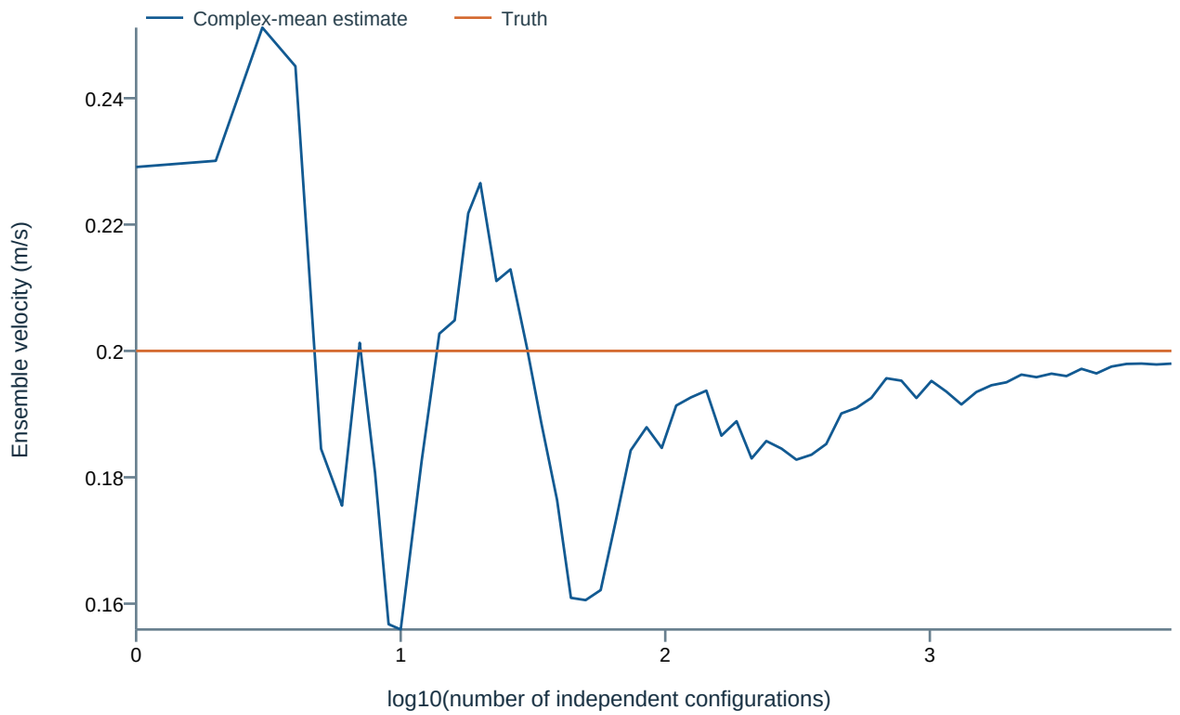
2 / Continuous scattering: verify the ensemble

Original 3-21 m continuous scene | 8192 independent configurations | TGC | No added receiver noise

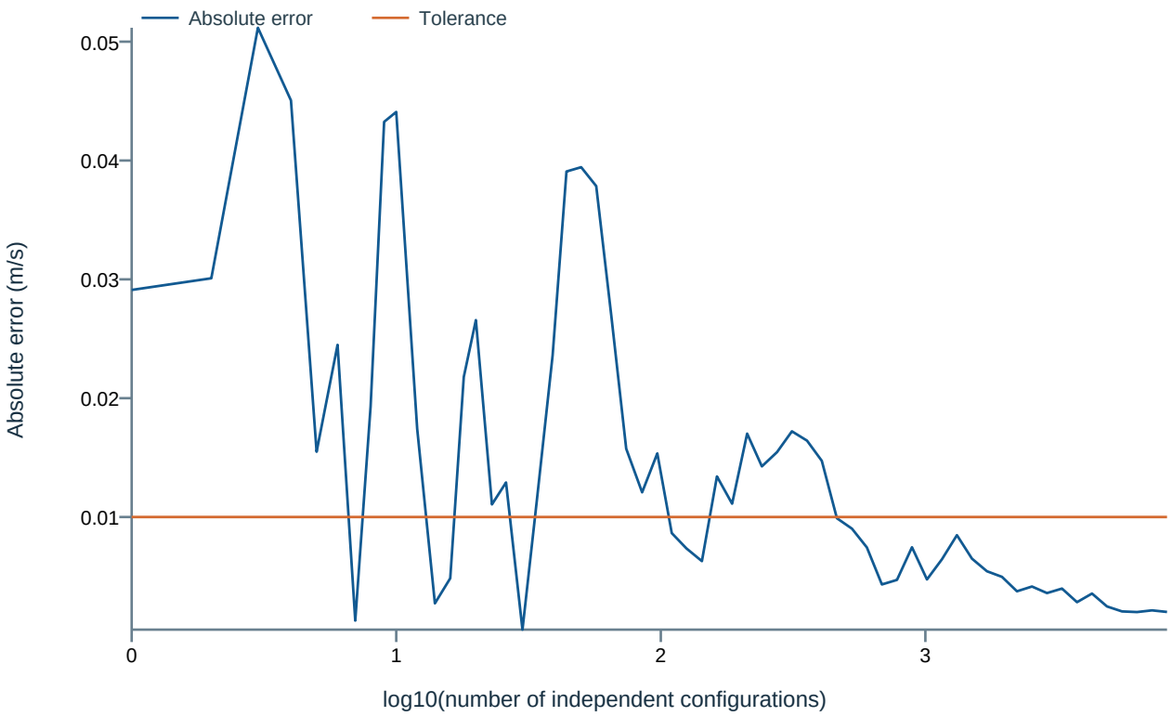
A. Individual estimates retain physical random error



B. Average correlations, then compute velocity



C. Final ensemble error is 0.00201 m/s



What each dataset validates

Controlled layers: RF mixing, FIR, delay alignment, gates, lag, velocity sign, scale and receive-time gain.

Continuous ensemble: random scattering and averaging. Fixed seeds; no selection of unusually accurate realizations.

Boundary +2.4 m/s: physical velocity is recovered.
Boundary +3.0 m/s: expected alias is verified and labelled.
A single lag cannot identify the true ambiguity interval.

The ensemble configurations are independent synthetic draws. They do not model real ping-to-ping time evolution.